

Luigi A. Moretti

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RESEARCH PROFILE

Luigi A. Moretti (he/him) is a medical doctor (MD) by training with a multidisciplinary bent. His career has been built on combining the skills and perspectives of medicine and technology. With hands-on involvement in the development of innovative digital products within startups and collaborations with various universities, he holds an MSc in Health Technology. Currently sponsored by the University of the West of England (UWE), Luigi is dedicated to co-designing an affective computing-based solution for mental health, specifically targeting anxiety disorders. His objective is to gather objective, interpretable and longitudinal data to support patient awareness and engagement, and to provide clinicians with an additional source of information to complement standard approaches such as self-report surveys.

EDUCATION

Doctor of Philosophy, School of Engineering, University of the West of England (UWE), Bristol, UK, April 2023 - in progress

- PhD project *“Integrating Affective Computing and Smart Sensing into Treatment Pathways for Anxiety Disorders”*, supervised by Dr David Western, UWE; Dr Miles Thompson, UWE; Dr Michael Loizou, University of Plymouth; Dr Paul Matthews, UWE.

Master of Science in Health Technology, University of the West of England (UWE), Bristol, UK, 2021-2022

- Final project *“Evaluating the translational potential of an emotion detection algorithm for healthcare applications”*, supervised by Dr David Western.
- Degree with Distinction.

6-year Degree in Medicine and Surgery, MD, Università di Bari Aldo Moro, Bari, Italy, graduated in 2020

- Experimental thesis in *“3D printing applications in Neurosurgery”*, supervised by Dr Francesco Signorelli, final mark 107/110.
- ERASMUS+ at Univerzita 'Pavla Jozefa Šafárika' v Košiciach, Kosice, Slovakia, Summer semester 2019.
- IFMSA Research Exchange at Université de Sfax, Tunisia, July 2018.

RESEARCH EXPERIENCE

Research Associate, UWE Bristol (UK): HUGxSmartSocks

September 2025 – in progress

- Ethics application
- Study protocol design
- Physiological data collection using wearables
- Qualitative and quantitative data analysis

PhD student, School of Engineering, University of the West of England (UWE), Bristol, UK,

April 2023 - in progress

- *Literature review and analysis.*
- *Research project planning and structuring.*
- *Relationship management with external organizations for implementing a co-design process.*
- *Project presentation in conferences and workshops.*
- *Facilitating co-design workshops and interviews, six phases so far with patients, carers, and clinicians.*
- *Prototype development (web app, mobile app, smartwatch app).*
- *Physiological data collection using wearables and their analysis.*

External researcher, Gastroenterology Department, Sapienza University, Rome, Italy, 2023 – 2024

- *3D segmentation and modelling of morphologically altered organs from CT scans.*
- *Discussion with stakeholders of the research aims and outcomes.*
- *Collaboration in future publications drafting for the methodology of the study.*

AI Health Start-up Co-founder, IntelliHearts, Italy, Bari, 2019-2021

- *Research about clinical implementation of an AI technology analyzing data collected by wearable devices (e.g. PPG, one-lead ECG) and developed within the rest of the team.*
- *Development of a UX and UI for a digital product (i.e. mobile app).*
- *Project presentation to investors, potential scientific and commercial partners (e.g. universities and businesses).*

Health Start-up Co-founder, MyHealthData (MyHD), Italy, Bari, 2018-2019

- *Research about pragmatic, secure and efficient ways to collect, store and share medical data both in daily and in emergency scenarios using wearable devices and NFC technology.*
- *Development of a UX and UI for a digital product (i.e. mobile app).*
- *Project presentation to public and scientific audiences.*

Digital manufacturing Start-up Co-founder, Climb3D (FabCraft Srls), Italy, Bari, 2017-2018

- *Research about eco-friendly, personalized, and light weight climbing tools and materials (e.g. climbing holds, wall panels, wood volumes, etc). Realised thanks to digital manufacturing techniques (e.g. 3D scanning, 3D modelling, 3D printing, CNC drilling).*
- *Prototypes reiterative development and products commercialization.*
- *Grant winning and small company management.*

GRANTS

- **NHS BNSSG ICB grant – in progress**, secured a £1,500 grant for PPIE activities from the NHS Bristol, North Somerset and South Gloucestershire Integrated Care Board.
- **CATE Public Engagement Fund (2025)**, Awarded an additional £3000 from UWE Bristol's '2024-25 CATE Public/Community Engagement & Knowledge Exchange Project Fund' to support the project's PPIE activities.
- **CATE Studentship by UWE, 2023 – in progress**, covering PhD Programme fees, up to £1.000 per year for research expenses and a monthly salary up to 3.5 years.
- **PassLaureati, 2021-2022**, public grant (up to €22.000) managed by Apulia Region (Italy) and financed by European Union (EU) to cover a Master's Degree Programme fees, and a monthly salary for the duration of the master.
- **Pugliesi Innovativi (PIN), 2018**, public grant (up to €30.000) managed by Apulia Region (Italy) and financed by European Union (EU) to develop an innovative project and a start-up launch: Climb3D (FabCraft Srls).

AWARDS

Won the SmartSocks Pioneers Competition by Milbotix (January 2026)

Best project in the 'Stress' category with the proposal named "Socks Over Stigma: Exploring SmartSocks as an Unobtrusive Alternative to Smartwatches for Emotion Monitoring in Anxiety Disorders – MEMoPAD."

Won Best Student Poster Award (June 2024)

'Best Student Poster' at the Second Workshop on Multimodal AI, organized by the University of Sheffield and the Alan Turing Institute.

International recognition and results achieved by IntelliHearts (since Luigi joined the team):

- *One of the top 10 projects selected: "COVID-19 Vaccine: Safety and Immunization Monitoring", organized by the Asian Development Bank (ADB) and Agorize.*
- *One of the top 15 projects in "Quarantine Hackathon" organized by IEEE sight Egypt.*
- *Semi-finalist Webit 2021.*
- *Letter of intent from Medical University of Lodz (PL) to perform clinical trials for IntelliHearts.*

StartCup Apulia 2019 winner: *MyHealthData (MyHD) has been considered one of the most valuable innovative projects in the Apulia region for the year 2019. This recognition granted access to present the project at the national competition between Italian Universities and Research Institutions called "Premio Nazionale innovazione", organized by PNI-Cube. Luigi was a co-creator and the team leader of MyHD.*

COMMUNITY INVOLVEMENT AND SERVICE

Oxfam Exchanger for Mediterranean Network Programme (MedNet), West Bank, Palestine, September 2019:

- Assisting in activities at Palestinian Incubator for Energy (PIE) an incubator for innovative projects, Jericho, Palestine
- Teaching Italian language at Birzeit University, Ramallah, Palestine

PUBLICATIONS

- L Moretti, Y Guan, Z Abdallah "Identifying emotion patterns in speech for Alzheimer's disease assessment: A novel perspective" *Behavioral Biometrics and Artificial Intelligence for Neurodegenerative* (2026). **[Published book chapter]**
- Moretti, Luigi A., et al. "MEMoPAD: A Co-Designed Wearable Emotion Monitoring System for Anxiety Disorders" *Interactive Health* (2026). **[Published conference paper]**
- Moretti, Luigi A., et al. "Digital Health Interventions Workshop" *CHI 2026, Digital Health Interventions Workshop* (2026). **[Published short paper - [link](#)]**
- Moretti, Luigi A., et al. "Co-Designing MEMOPAD: Multimodal Emotion Monitoring in clinical Pathways for Anxiety Disorders." *BSC HCI 2025 Workshop* (2025). **[Published Position Paper - [link](#)]**
- Liu, Xianyan, et al. "Towards deployment-centric multimodal AI beyond vision and language." *Nature Machine Intelligence* (2025): 1-13. **[Published Journal Paper]**
- Draper, Thomas C., et al. "Clinical AI Scribes in primary care: accuracy, error severity and implications for clinical practice." *BMJ Digital Health & AI* 1.1 (2025). **[Published Journal Paper]**
- Moretti, Luigi A., et al. "Affective Computing in Anxiety Disorders: A Rapid Literature Review of Emotion Recognition Applications." *Proceedings of the 18th International Joint Conference on Biomedical Engineering Systems and Technologies*. 2025. **[Published Conference Paper]**
- Moretti, L.A., Thompson, M., Loizou, M., Matthews, P., Western, D. (2024). *Co-Designing an Affective Computing Solution for Ecological momentary assessment in Anxiety Disorders*. **[Conference Abstract to the 2nd International Digital Mental Health and Wellbeing Conference]**
- Dentamaro, V., Impedovo, D., Moretti, L.A., Pirlo, G., & Suresh, P.K. (2023). *An Approach using transformer architecture for emotion recognition through Electrocardiogram Signal(s)*. *DSTNDS*. **[Published Conference Paper]**
- Dentamaro, V., Giglio, P., Impedovo, D., Moretti, L.A., Pirlo, G., & Sblendorio, E. (2023). *A benchmarking study of deep learning techniques applied for breath analysis*.

DSTNDS. **[Published Conference Paper]**

- Dentamaro, V., Giglio, P., Impedovo, D., Moretti, L.A., & Pirlo, G. (2022). *AUCO ResNet: an end-to-end network for Covid-19 pre-screening from cough and breath*. *Pattern Recognition*, 127, 108656 - 108656. **[Published Journal Paper]**

REFERENCES

David Western, PhD

*Wallscount Fellow (Senior Lecturer) in Health Technology,
Leader of the [REACH Research Group](#)*

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